

Exercise J01

Q1. The value of the determinant $\begin{vmatrix} a^2 & a & 1 \\ \cos(nx) & \cos(n+1)x & \cos(n+2)x \\ \sin(nx) & \sin(n+1)x & \sin(n+2)x \end{vmatrix}$ is independent of :

- (A) n (B) a (C) x (D) a, n and x

Ans. A

Sol. [Hint: Directly open by R_1 to get a form of $\sin(A-B)$ etc.]

Q2. If a, b, c are all different from zero & $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$, then the value of $a^{-1} + b^{-1} + c^{-1}$ is

- (A) abc (B) $a^{-1} b^{-1} c^{-1}$ (C) $-a - b - c$ (D) -1

Ans. D

Sol. [Hint : $C_1 \rightarrow C_1 - C_2$ & $C_2 \rightarrow C_2 - C_3$ & then open by R_1 to get $ab + abc + ac + bc = 0$; divided by abc]

Q3. If α, β & γ are the roots of the equation $x^3 + px + q = 0$, then the value of the determinant $= \begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$

- (A) p (B) q (C) $p^2 - 2q$ (D) none

Ans. D

Sol. $D = \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma$

Suppose given $\alpha + \beta + \gamma = 0$

as coefficient of $x^2 = 0$

Now $\det(\alpha + \beta + \gamma) \begin{vmatrix} 1 & 1 & 1 \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0 \Rightarrow (D)$

Q4. If $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$ term of a GP are a, b, c then the value of $\begin{vmatrix} \log l & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix}$ is equal to-

- (A) 0 (B) 1 (C) $l + m + n$ (D) None of these

Ans. A

Sol.
$$\begin{vmatrix} \log l & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix} = 0$$

Here $a = AR^{p-1}$, $B = AR^{q-1}$, $C = AR^{r-1}$

$$\begin{vmatrix} \log AR^{p-1} & p & 1 \\ \log AR^{q-1} & q & 1 \\ \log AR^{r-1} & r & 1 \end{vmatrix} = \begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log A + (q-1)\log R & q & 1 \\ \log A + (r-1)\log R & r & 1 \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1$ & $R_3 \rightarrow R_3 - R_1$

$$\begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log R(q-p) & q-p & 0 \\ \log R(r-p) & r-p & 0 \end{vmatrix} = (r-p)(q-p) \begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log R & 1 & 0 \\ \log R & 1 & 0 \end{vmatrix} = 0$$

Q5. If $a_1, a_2, a_3, \dots, a_n, \dots$ are in G.P., then the value of the determinant

$$\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}, \text{ is}$$

(A) 0 (B) 1 (C) 2 (D) -2

Ans. A

Sol. If $a_1, a_2, \dots, a_n, \dots$ are in G.P.

then $\log a_1, \log a_2, \dots, \log a_n, \dots$ are in A.P.

A, A + D,

Let common difference of A.P. is D

$$\text{so } \begin{vmatrix} \log a_n & \log a_{n+2} & \log a_{n+4} \\ \log a_{n+6} & \log a_{n+8} & \log a_{n+10} \\ \log a_{n+12} & \log a_{n+14} & \log a_{n+16} \end{vmatrix}$$

$$\begin{vmatrix} A & A+2D & A+4D \\ A+6D & A+8D & A+10D \\ A+14D & A+12D & A+16D \end{vmatrix}$$

$C_2 \rightarrow 2C_2 - (C_1 + C_3)$

$$\begin{vmatrix} A & 0 & A+4D \\ A+6D & 0 & A+10D \\ A+12D & 0 & A+16D \end{vmatrix} = 0$$

Q6. If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{bmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{bmatrix}$ then $f(x)$ is a polynomial of degree

- (A) 1 (B) 0 (C) 3 (D) 2

Ans. D

Sol. On applying $C_1 \rightarrow C_1 + C_2 + C_3$

$$\text{we } \begin{bmatrix} 1 & x + b^2x & c^2x + x \\ 1 & 1 + b^2x & c^2x + x \\ 1 & x + b^2x & c^2x + 1 \end{bmatrix} \mathbf{x} = (\mathbf{x})$$

Now solve it.

Q7. If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0, y \neq 0$ then D is-

- (A) divisible by neither x nor y (B) divisible by both x and y (C) divisible by x but not y
(D) divisible by y but not x

Ans. B

Sol. $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$

$$C_1 \rightarrow C_1 - C_2, C_2 \rightarrow C_2 - C_3$$

$$\begin{vmatrix} 0 & 0 & 1 \\ -x & x & 1 \\ 0 & -y & 1+y \end{vmatrix} = xy = xy$$

$\therefore$ D is divisible by both x and y

Q8. Let a, b, c be such that $b(a+c) \neq 0$. If

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^nc \end{vmatrix} = 0, \text{ then the value of n is :}$$

- (A) any even integer (B) any odd integer (C) any integer (D) zero

Ans. B

Sol. $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ a & -b & c \end{vmatrix} = \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & a-1 & a \\ b+1 & b-1 & -b \\ c-1 & c+1 & c \end{vmatrix}$

$$= \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^{n+1} \begin{vmatrix} a+1 & a & a-1 \\ b+1 & -b & b-1 \\ c-1 & c & c+1 \end{vmatrix}$$

$$C_2 \longleftrightarrow C_3$$

$$= (1 + (-1)^{n+2}) \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix}$$

This is equal to zero only if $n + 2$ is odd i.e. n is odd integer.

Q9.

If $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, then Δ^2 is equal to -

$$(A) \begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix} \quad (B) \begin{vmatrix} bc+a^2 & ca+b^2 & ab+c^2 \\ ca+b^2 & ab+c^2 & bc+a^2 \\ ab+c^2 & bc+a^2 & ca+b^2 \end{vmatrix} \quad (C) \begin{vmatrix} a^2-bc & b^2-ca & c^2-ab \\ b^2-ca & c^2-ab & a^2-bc \\ c^2-ab & a^2-bc & b^2-ca \end{vmatrix}$$

(D) None of these

Ans. A

Sol. -

Q10.

If $ax + by + cz = 1$, $bx + cy + az = 0 = cx + ay + bz$, then $\begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix} \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ is equal to -

(A) 0 (B) 1 (C) -1 (D) 2

Ans. B

Sol. -

Q11. If $\alpha, \beta \neq 0$ and $f(n) = \alpha^n + \beta^n$ and

$$\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1-\alpha)^2 (1-\beta)^2 (\alpha-\beta)^2 \text{ then } K \text{ is equal to :}$$

- (A) 1 (B) -1 (C) $\alpha\beta$ (D) $1/\alpha\beta$

Ans. A

Sol.
$$\begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{bmatrix}^2$$

$$= (1-\alpha)^2 (1-\beta)^2 (\alpha-\beta)^2$$

Q12. Number of value of 'a' for which the system of equations,

$$a^2 x + (2-a)y = 4+a^2$$

$$a x + (2a-1)y = a^5 - 2 \text{ possess no solution is}$$

- (A) 0 (B) 1 (C) 2 (D) infinite

Ans. C

Sol.
$$\frac{a^2}{a} = \frac{2-a}{2a-1} \neq \frac{4+a^2}{a^5-2} \{-1, 1\} \text{ Ans.}$$

Q13. The following system of equations $3x - 7y + 5z = 3$; $3x + y + 5z = 7$ and $2x + 3y + 5z = 5$ are

- (A) consistent with trivial solution (B) consistent with unique non trivial solution
(C) consistent with infinite solution (D) inconsistent with no solution

Ans. B

Sol. $D \neq 0 \Rightarrow \text{consistency}$

Q14. If the system of equations

$$x - 2y + z = a$$

$$2x + y - 2z = b$$

$$\text{and } x + 3y - 3z = c$$

have atleast one solution, then the relationship between a, b and c is

- (A) $a + b + c = 0$ (B) $a - b + c = 0$ (C) $-a + b + c = 0$ (D) $a + b - c = 0$

Ans. B

Sol.
$$D = \begin{vmatrix} a & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 3 & -3 \end{vmatrix} \text{ which vanishes; hence for atleast one solution } D_1 = D_2 = D_3 = 0$$

$$\therefore D_1 = \begin{vmatrix} a & -2 & 1 \\ b & 1 & -2 \\ c & 3 & -3 \end{vmatrix} = 0 \Rightarrow a - b + c = 0 \text{ Ans.}$$

Q15. Let α, β, γ are the real roots of the equation $x^3 + ax^2 + bx + c = 0$ ($a, b, c \in \mathbb{R}$ and $a \neq 0$).

If the system of equations (in u, v and w) given by

$$\alpha u + \beta v + \gamma w = 0$$

$$\beta u + \gamma v + \alpha w = 0$$

$$\gamma u + \alpha v + \beta w = 0$$

has non-trivial solutions, then a^2 equals

(A) b (B) $2b$ (C) $3b$ (D) $4b$

Ans. C

Sol.

$$\text{We have } x^3 + ax^2 + bx + c = 0 \quad \begin{matrix} \alpha \\ \beta \\ \gamma \end{matrix}$$

$$\therefore \alpha + \beta + \gamma = -a$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = b$$

Since the given system of equations has non-trivial solutions, so

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0$$

$$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma = 0 \quad \Rightarrow (\alpha + \beta + \gamma) [\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha] = 0$$

$$\Rightarrow (\alpha + \beta + \gamma) [(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)] = 0$$

$$\Rightarrow -a [a^2 - 3b] = 0$$

Hence $a^2 = 3b$ (as $a \neq 0$)

Q16. If the system of linear equations

$$x + 2ay + az = 0$$

$$x + 3by + bz = 0$$

$$x + 4cy + cz = 0$$

has a non-zero solution, then a, b, c

(A) are in G.P. (B) are in H.P. (C) satisfy $a + 2b + 3c = 0$ (D) are in A.P.

Ans. B

Sol. For non zero solution

$$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$$

Q17.

Ans. -

Sol. -

Q18. The existence of unique solution of the system $x + y + z = b$, $2x + 3y - z = 6$, $5x - y + az = 10$ depends on —

(A) b only (B) a only (C) a and b (D) neither a nor b

Ans. B

Sol. -

Q19. The system equations $\alpha x + y + z = \alpha - 1$, $x + \alpha y + z = \alpha - 1$, $x + y + \alpha z = \alpha - 1$ has no solution, if α is -

(A) -2 (B) either -2 or 1 (C) not -2 (D) 1

Ans. A

Sol. For no solution

$\Delta = 0$ and Δ_x or Δ_y or Δ_z at least one is not zero.

$$\Delta = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = 0$$

at $\alpha = 1$ their are 3 row are identical so factor of determinant $(\alpha - 1)^2 = 0$

$$\lambda(\alpha + 2)(\alpha - 1)^2 = 0$$

$$\alpha = -2, 1$$

but at $\alpha = 1$

all equation are same so at $\alpha = 1$ system of equation infinite solution and at $\alpha = -2$

$$\Delta_x = \begin{vmatrix} -3 & 1 & 1 \\ -3 & -2 & 1 \\ -3 & 1 & -2 \end{vmatrix}$$

$$-3(4 - 1) - 1(6 + 3) + 1(-3 + 6)$$

$$-9 - 9 + 3 = -15 \neq 0$$

so at $\alpha = -2$ system have no solution

Q20. The number of values of k , for which the system of equations :-

$$(k + 1)x + 8y = 4k,$$

$$kx + (k + 3)y = 3k - 1$$

has no solution, is

- (A) infinite (B) 1 (C) 2 (D) 3

Ans. B

Sol. $\frac{k+1}{8} = \frac{8}{k+3} \neq \frac{4k}{3k-1} \Rightarrow k^2 + 4k + 3 = 8k \Rightarrow k^2 - 4k + 3 = 0$

$$k = 1, 3$$

If $k = 1$ then $\frac{8}{1+3} \neq \frac{4 \cdot 1}{2}$ False

And If $k = 3$ then $\frac{8}{6} \neq \frac{4 \cdot 1}{2}$

therefore $k = 3$ / Hence only one value of k .

Q21. The set of all values of λ for which the system of linear equation :

$$2x_1 - 2x_2 + x_3 = \lambda x_1$$

$$2x_1 - 3x_2 + 2x_3 = \lambda x_2$$

$$-x_1 + 2x_2 = \lambda x_3 \quad \text{has a non-trivial solution,}$$

- (A) is an empty set (B) is a singleton (C) contains two elements
(D) contains more than two elements

Ans. C

Sol.
$$\begin{vmatrix} 2-\lambda & -2 & 1 \\ 2 & -3-\lambda & 2 \\ -1 & 2 & -\lambda \end{vmatrix}$$

Q22. If S is the set of distinct values of ' b ' for which the following system of linear equations

$$x + y + z = 1$$

$$x + ay + z = 1$$

$$ax + by + z = 0$$

has no solution, then S is

- (A) a finite set containing two or more elements (B) a singleton (C) an empty set
(D) an infinite set

Ans. B

Sol.
$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & a & 1 \\ a & b & 1 \end{vmatrix} = -(a-1)^2 = 0$$

$$a = 1$$

If $a = 1$ two planes are parallel

So for no solution

Third plane must be parallel

$$b = 1$$